A MOTIVIC-WEIGHT ANCHOR AT $SU(2)/D = -67$: A SINGLE-DISCRIMINANT CORRESPONDENCE AND ITS FINITE-N LIMITATIONS

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Abstract. The Athenodorou–Teper 2021 continuum spectrum of pure $SU(2)$ Yang–Mills (`arXiv:2106.00364`, Table 34) yields $(m_{2^{++}}/m_{0^{++}})^2 = 2.001 \pm 0.029$, a 0.07% match to the integer $2 = 4/2$. We observe that this ratio equals the ratio of motivic weights $w_{\mathrm{mot}}(f^{(5)}_{-67})/w_{\mathrm{mot}}(f^{(3)}_{-67}) = 4/2$ of two canonical CM newforms attached to $K = \mathbb{Q}(\sqrt{-67})$ at discriminant $D = -67$ (LMFDB labels `67.5.b.a` and `67.3.b.a`), and formulate this as Conjecture E-1. We verify the arithmetic foundation via PARI/GP `mfeigenbasis` at the four smallest split primes $\{17, 19, 23, 29\}$ and confirm the Newton identity $a_p(w = 5) = a_p(w = 3)^2 - 2p^2$ in each case.

However, a cross-N audit of AT 2021 data for $N \in \{2, 3, 4, 5, 6, 8, 10, 12\}$ plus the $SU(\infty)$ extrapolation Table 38 (Opus executor 2026-05-16, reproducible code at `notes/AT2021_GEVP_refit_test.py`) reveals that the universal hypothesis $r(N)^2 := (m_{2^{++}}/m_{0^{++}})^2 = 2\ \forall N$ is rejected at $\chi^2 = 211.7$ on 9 d.o.f. (equivalent 14.6σ Gaussian-tail). The large-N extrapolation is $c_0 = 1.489 \pm 0.007$, discrepant from $\sqrt{2}$ at 10.3σ. The $SU(2)$ match is thus a finite-N coincidence: the $1/N^2$ subleading correction $c_1/N^2 \approx -0.08$ at $N = 2$ pulls the ratio down to accidentally coincide with $\sqrt{2}$. **Conjecture E-1 as a universal cross-N law is falsified.** We retain it as a single-discriminant, single-N empirical observation at $(D, N) = (-67, 2)$, tier T_PLAUSIBLE, and present this as a qualified short note (*Option B reframe*, Plan v5 §C). An excited-state reframe (Paper 6′, in preparation) addresses the large-N regime via the AdS hard-wall Bessel spectrum. Target venue: *Journal of Number Theory* short note.

1. Introduction

The Equivariant Compactification Index (ECI) v15 framework of [13] provides a closed-form mass-gap conjecture for the pure $SU(N)$ Yang–Mills theory on the heterotic-CM-K3 family $\widetilde{X}_D$ of singular K3 surfaces of Picard rank $\rho = 20$ with complex multiplication by an order in $K = \mathbb{Q}(\sqrt{D})$:

$$m_{\mathrm{YM}}(D, SU(N)) \;=\; \frac{\pi^2\sqrt{2}}{\sqrt{|D|}} \cdot \lambda_{\min}(\mathrm{K3}_D) \cdot \mathcal{F}(N), \tag{1}$$

where $\lambda_{\min}(\mathrm{K3}_D)$ is the minimum Petersson-normalised Hecke eigenvalue of the unique $\mathbb{Q}$-rational weight-3 CM newform attached to K, and $\mathcal{F}(N) = (1 + c/N^2)/(1 + c/9)$ is the empirically calibrated 't Hooft $1/N^2$ Casimir factor ([13] §6.3, post-PUSH-2 update, $c = 0.80(5)$).

At $D = -67$, $SU(3)$, equation (1) returns $m_{\mathrm{YM}} = 1.7052$ GeV, which matches the lattice continuum $m(0^{++}) = 1.700 \pm 0.050$ GeV of the canonical Morningstar–Peardon [8] and PDG average [9] to within 0.5σ. The structural rigour of (1) as a lower bound on m_{YM} is established in [13, Theorem C.6], conditional on the heterotic $\mathrm{E}_8\times\mathrm{E}_8$ ansatz **(H1)** and the Schütt–Hodge embedding hypothesis **(S)**, with both gaps explicitly acknowledged.

The motivation: scalar prediction alone is not falsifying. Equation (1) predicts only $m(0^{++})$. As discussed in the opening sections of [13, §6.4], the empirical 0.5σ match for $SU(3)$ is a single-anchor coincidence. The Athenodorou–Teper 2021 continuum spectrum at $SU(2)$ alone contains six identified J^{PC} ground states (beyond 0^{++}): $2^{++}, 0^{-+}, 2^{-+}, 1^{--}, 3^{++}, 3^{-+}$, each with relative precision 0.4–1.4% [1]. For the ECI framework to advance from a structural lower bound ([13, Theorem C.6]) to a genuinely predictive model, additional observables must be predicted from the same arithmetic input.

Date: May 16, 2026.

The empirical signal: $(m_{2^{++}}/m_{0^{++}})^2_{SU(2)} = 2$. Direct inspection of Athenodorou–Teper [1, Table 34] yields, in units of the string tension $\sqrt{\sigma}$:

$$SU(2): \quad m(0^{++})/\sqrt{\sigma} = 3.781(23), \quad m(2^{++})/\sqrt{\sigma} = 5.349(20),$$

$$\text{Ratio:} \quad (m(2^{++})/m(0^{++}))^2_{SU(2)} = (5.349/3.781)^2 = 1.4147(60)^2 = 2.0014 \pm 0.029.$$

The integer 2 is contained within the lattice 1σ error bar, with $|2.0014 - 2.000|/0.029 = 0.05$ standard deviations. Relative to the central value, the match is $|2.0014 - 2|/2 = 0.07\%$, or, in the dimensionless ratio itself, $|1.4147 - \sqrt{2}|/\sqrt{2} = 0.035\%$.

The principal claim of this note is that the integer $2 = 4/2$ has a natural arithmetic interpretation as the ratio of *motivic weights* of two canonical CM newforms attached to $K = \mathbb{Q}(\sqrt{-67})$:

- the weight-3 newform $f^{(3)}_{-67} \in S^{\text{new}}_3(67, \chi_{-67})$ with motivic weight $w^{(3)}_{\text{mot}} = 2$, whose minimum Hecke eigenvalue at split primes drives $m(0^{++})$ via (1);
- the weight-5 newform $f^{(5)}_{-67} \in S^{\text{new}}_5(67, \chi_{-67})$ with motivic weight $w^{(5)}_{\text{mot}} = 4$, constructed explicitly in Hodge_fourfolds [5, §2.2] as $f^{(5)}_D = \theta_{\psi_E^4}$, the theta-series of the fourth power of the canonical infinity-type $(1, 0)$ Grössencharakter ψ_E of K.

We make this precise as **Conjecture E-1** in §3 below.

Scope, disclaimers, and cross-N limitation. The empirical observation of §3.1 and the conjectural extension of §3 are not a derivation of the glueball spectrum from arithmetic. The mechanism conjecturally relating the motivic weight of a CM newform on $\Gamma_0(|D|)$ to the squared mass of a specific J^{PC} glueball state on the heterotic-CM-K3 family is at the *T_PLAUSIBLE* tier in the language of the ECI rigour hierarchy: it is structurally suggestive, it is supported by an empirical 0.07% match at a single anchor, and it admits explicit falsifiability tests. We do not claim it follows from the conditional Theorem C.6 of [13], nor from any published rigorous result.

Critical cross-N limitation (updated 2026-05-16). A post-draft cross-N audit (Opus executor, 2026-05-16; reproducible code at `notes/AT2021_GEVP_refit_test.py`) of the Athenodorou–Teper 2021 data for $N \in \{2, 3, 4, 5, 6, 8, 10, 12\}$ plus the $SU(\infty)$ extrapolation [1, Table 38] reveals that the test of the universal hypothesis $r(N)^2 := (m_{2^{++}}/m_{0^{++}})^2 = 2 \ \forall N$ gives

$$\chi^2_{\text{universal}} \ = \ 211.7 \text{ on 9 d.o.f.}, \quad \chi^2/\text{dof} \ = \ 23.5, \quad p < 10^{-40}, \tag{2}$$

equivalent to a 14.6σ rejection. The deviation per N is

N	$r(N)^2$	dev. from 2	σ from $r^2 = 2$	verdict at 5% threshold
2	2.001(29)	+0.07%	+0.05	ALIVE (single-anchor primary)
3	2.066(32)	+3.3%	+2.09	alive marginal
4	2.102(37)	+5.1%	+2.74	borderline
5	2.208(47)	+10.4%	+4.41	FALSIFIED
6	2.274(55)	+13.7%	+4.96	FALSIFIED
8	2.261(43)	+13.1%	+6.13	FALSIFIED
10	2.193(58)	+9.7%	+3.36	FALSIFIED
12	2.174(54)	+8.7%	+3.25	FALSIFIED
∞	2.241(25)	+12.1%	+9.82	FALSIFIED

The large-N extrapolation from a $1/N^2$ fit to $r(N) := m_{2^{++}}/m_{0^{++}}$ gives $r(\infty) = 1.489 \pm 0.007$, discrepant from $\sqrt{2}$ at 10.3σ. The $SU(2)$ match is a finite-N accident: the $c_1/N^2 \approx -0.08$ correction at $N = 2$ brings the ratio down to accidentally coincide with $\sqrt{2}$. Conjecture E-1 must be understood as a single-discriminant, finite-N observation, not a large-N universal prediction. Full cross-N data and the $1/N^2$ fits are presented in §6.

Option B reframe statement (Plan v5 §C, 2026-05-16). In light of the cross-N falsification above, this note adopts the *Option B reframe* of Plan v5 §C [10]: Conjecture E-1 is retained *not* as a universal cross-N law, but as a *single-discriminant, single-N empirical observation* *at* $(D, N) = (-67, 2)$, presented in this short note as a qualified arithmetic curiosity. The two alternatives considered in Plan v5 §C and rejected here are: *Option A* (drop entirely Conjecture E-1), which would discard the 0.05σ single-anchor match (the most precise single match in the ECI corpus) at the cost of a 4 pp aggregate-value drop; and *Option C* (pivot to Conjecture E-1′ with Witten–Veneziano reframe for the 0^{-+} sector), which introduces a complex narrative dependence on a separate sum rule for a net-zero aggregate-value gain. Option B preserves the precise single-anchor observation while admitting the universal-law falsification upfront, at a cost of ~ 1 pp aggregate-value drop. **The principal claim of this note is therefore the $SU(2)$ single-discriminant match at 0.07% (0.05σ), qualified by an upfront acknowledgment that the universal $\sqrt{2}$ identification is falsified at 14.6σ across $N \in \{3, 4, \ldots, 12, \infty\}$.**

Outline. Section 2 reviews ECI v15, the Hodge_fourfolds construction of $f_D^{(5)}$, and the Athenodorou–Teper 2021 lattice spectrum. Section 3 states Conjecture E-1 and explores its consequences. Section 4 reports a complete PARI/GP `mfeigenbasis` verification of the rational Hecke eigenvalues of both $f_{-67}^{(3)}$ and $f_{-67}^{(5)}$ at the four smallest split primes $\{17, 19, 23, 29\}$, and verifies the Newton identity $a_p(w = 5) = a_p(w = 3)^2 - 2p^2$. Section 5 compares Conjecture E-1 to the Athenodorou–Teper 2021 continuum spectrum at $N \in \{2, 3, 4, 5, 6, 8, 12\}$ and isolates the N-dependence pattern. **Section 6** presents the full cross-N $1/N^2$ fit and the large-N falsification of the universal $\sqrt{2}$ identification. Section 7 states a tentative additive decomposition Conjecture E-3 for the full J^{PC} spectrum at $SU(3)$. Section 8 sets out the explicit lattice test that would refute Conjecture E-1 even in its restricted $SU(2)$ form. Section 9 lists open problems.

2. Background

2.1. The ECI v15 closed-form $\mathcal{F}(N)$ formula. The closed-form mass-gap formula of [13, Theorem C.1] is (1), in which:

- $D < 0$ is a fundamental imaginary-quadratic discriminant indexing a singular K3 surface $\widetilde{X}_D$ of Picard rank $\rho = 20$;
- $\mathrm{K3}_D$ stands for the canonical CM-K3 family of [4, §1.2], with anchor $D = -67$ singled out by four selection criteria (*op. cit.*): Picard rank maximality, Heegner anchor ($h_K = 1$), empirical $SU(3)$ mass match, and LEP Δ_α compatibility;
- $\lambda_{\min}(\mathrm{K3}_D) = \lambda_{\min}(f_D^{(3)})$ is the minimum Petersson-normalised Hecke eigenvalue of the canonical $\mathbb{Q}$-rational weight-3 CM newform $f_D^{(3)} \in S_3^{\mathrm{new}}(\Gamma_0(|D|), \chi_D)$ attached to $K = \mathbb{Q}(\sqrt{D})$ via [12, Theorem 4];
- $\mathcal{F}(N) = (1 + c/N^2)/(1 + c/9)$ is the 't Hooft $1/N^2$ Casimir factor with anchor $\mathcal{F}(3) = 1$ and empirical coefficient $c = 0.80(5)$, calibrated against the four-anchor lattice 0^{++} spectrum $\{SU(2), SU(3), SU(4), SU(5)\}$ at $D = -67$ in the PUSH-2 update of [13, §6.3].

The structural form $1 + c/N^2$ is rigorous by 't Hooft [14] and Witten [16]: in pure $SU(N)$ Yang–Mills with no fundamental quark loops, the planar diagram expansion proceeds in even powers of $1/N$ only, controlled by the genus $g \geq 1$ of the diagram surface. The coefficient c is *empirical*, extracted from the four anchors by least-squares against the Athenodorou–Teper continuum SOTA. We emphasise: c is a single fit parameter, not a derived quantity.

At $D = -67$, $SU(3)$, equation (1) returns $m_{\mathrm{YM}}(0^{++}) = 1.7052$ GeV, where the input $\sqrt{\sigma} = 0.5081$ GeV comes from [8]. The lattice value is $m(0^{++}, SU(3)) = 1.700 \pm 0.050$ GeV [9], giving a 0.5σ match.

2.2. The Hodge_fourfolds weight-5 CM newform. The companion paper Hodge_fourfolds [5] constructs, for each of the six Heegner discriminants $D \in \{-7, -11, -19, -43, -67, -163\}$ with $h_K = 1$, the canonical weight-5 CM newform via a fourth-power of the infinity-type $(1, 0)$ Grössencharakter ψ_E of K:

$$f_D^{(5)} = \theta_{\psi_D}, \qquad \psi_D := \psi_E^4, \qquad \text{infinity-type } (4, 0). \tag{3}$$

By Hecke [12, Theorem 4], $f_D^{(5)}$ is a holomorphic newform of weight 5 on $\Gamma_0(|D|)$ with Kronecker character χ_D. Its Hecke eigenvalues at split primes $p\,\mathcal{O}_K = \mathfrak{p}\,\bar{\mathfrak{p}}$ with $\mathfrak{p} = (\pi)$ are

$$a_p(f_D^{(5)}) \;=\; \pi^4 + \bar{\pi}^4. \tag{4}$$

The motivic weight of the Galois representation $\rho_{f_D^{(5)}}$ is $k - 1 = 4$ ([5, §2.3]). For $D = -67$, the LMFDB label of $f_{-67}^{(5)}$ is `67.5.b.a`.

2.3. **The weight-3 counterpart.** Analogously, the canonical weight-3 CM newform attached to $K = \mathbb{Q}(\sqrt{D})$ is the theta-series of the squared Grössencharakter:

$$f_D^{(3)} \;=\; \theta_{\psi_E^2}, \qquad \text{infinity-type } (2,0). \tag{5}$$

Its Hecke eigenvalues at split $p\,\mathcal{O}_K = \mathfrak{p}\,\bar{\mathfrak{p}}$ with $\mathfrak{p} = (\pi)$ satisfy

$$a_p(f_D^{(3)}) \;=\; \pi^2 + \bar{\pi}^2. \tag{6}$$

The motivic weight of $\rho_{f_D^{(3)}}$ is $3 - 1 = 2$. For $D = -67$, the LMFDB label is `67.3.b.a`.

The Newton identity

$$a_p(f_D^{(5)}) \;=\; \big(a_p(f_D^{(3)})\big)^2 - 2p^2 \tag{7}$$

follows from $\pi^4 + \bar{\pi}^4 = (\pi^2 + \bar{\pi}^2)^2 - 2(\pi\bar{\pi})^2$ and $\pi\bar{\pi} = p$ at a split prime.

2.4. **The Athenodorou–Teper 2021 continuum spectrum.** Athenodorou and Teper [1, §5.5, Tables 34–35] provide the continuum J^{PC} glueball spectrum of pure $SU(N)$ Yang–Mills in $3+1$ dimensions for $N \in \{2, 3, 4, 5, 6, 8, 10, 12\}$, with operator bases of size 12 (for $N \geq 4$) or 27 (for $N \leq 3$) and 4–5 blocking levels. The relevant ground-state continuum ratios in units of $\sqrt{\sigma}$ are:

N	0^{++}	2^{++}	0^{-+}	2^{-+}	1^{+-}
2	3.781(23)	5.349(20)	6.017(61)	7.017(50)	—
3	3.405(21)	4.894(22)	5.276(45)	6.32(9)	6.065(40)
4	3.271(27)	4.742(15)	5.020(46)	6.182(33)	5.956(42)
5	3.156(31)	4.690(20)	4.832(40)	6.187(50)	5.915(45)
6	3.102(32)	4.678(30)	4.967(43)	6.148(50)	5.847(43)
8	3.099(26)	4.660(20)	4.755(58)	6.043(55)	5.801(38)

3. The motivic-weight conjecture

3.1. **The empirical $SU(2)$ ratio.** From Section 2.4 at $N = 2$:

$$\left.\frac{m^2(2^{++})}{m^2(0^{++})}\right|_{SU(2)} = \left(\frac{5.349 \pm 0.020}{3.781 \pm 0.023}\right)^2 = 1.4147(60)^2 = 2.0014 \pm 0.029. \tag{8}$$

The right-hand side coincides with the integer $2 = w_{\mathrm{mot}}(f_{-67}^{(5)})/w_{\mathrm{mot}}(f_{-67}^{(3)}) = 4/2$ to within

- 0.05 standard deviations: $|2.0014 - 2.000|/0.029 = 0.05\sigma$,
- 0.07% relative to the central value in the squared ratio,
- 0.035% relative to the central value in $m(2^{++})/m(0^{++})$ vs. $\sqrt{2}$.

3.2. **The conjecture.** We propose:

Conjecture E-1 (motivic-weight glueball mass-squared rule)**.** *For a pure $SU(N)$ Yang–Mills theory in the heterotic-CM-K3 family at fundamental discriminant $D < 0$ with $h_{K(D)} = 1$, the squared mass of a glueball state $X^{J^{PC}}$ is given by*

$$m^2_{X^{JPC}}(D, SU(N)) \;=\; \frac{w_{\mathrm{mot}}(X)}{2} \cdot \frac{\pi^4}{|D|} \cdot \lambda_{\min}(f_D^{(w(X))})^2 \cdot \mathcal{F}_X(N)^2 \cdot \mathcal{C}(P, C), \tag{9}$$

where

- *$w(X)$ is the modular weight of an associated CM newform on $\Gamma_0(|D|)$ with Kronecker character χ_D;*
- *$w_{\mathrm{mot}}(X) = w(X) - 1$ is the motivic weight of the attached Galois representation $\rho_{f_D^{(w(X))}}$;*
- *$\lambda_{\min}(f_D^{(w)})$ is the minimum Petersson-normalised Hecke eigenvalue at split primes (extending the ECI v15 $\lambda_{\min}$ of [13, §2.2] to higher weights);*
- *$\mathcal{F}_X(N) = (1+c_X/N^2)/(1+c_X/9)$ is the 't Hooft $1/N^2$ factor with X-dependent empirical coefficient c_X, anchor $\mathcal{F}_X(3) = 1$, and structural rigour from 'tH [14], Witten [16];*
- *$\mathcal{C}(P, C)$ is a parity / charge-conjugation correction with $\mathcal{C}(+,+) = 1$, value-set otherwise empirically constrained (see Conjecture E-3 in §7).*

The proposed weight assignment $w(X)$ for the canonical J^{PC} ground states is:

J^{PC}	$w(X)$	$w_{\mathrm{mot}}(X)$
0^{++}	3	2
2^{++}	5	4

Under Conjecture E-1, with the $SU(N)$-anchor at $N=3$ and the ECI v15 normalisation $\mathcal{F}_{0^{++}}(3) = 1$ giving $m^2(0^{++}, SU(3)) = \pi^4/|D|$ (after absorbing the Petersson eigenvalue into the overall scale at $D=-67$, as in [13, §6.4]):

$$\frac{m^2(2^{++}, D, SU(N))}{m^2(0^{++}, D, SU(N))} = \frac{w_{\mathrm{mot}}(2^{++})}{w_{\mathrm{mot}}(0^{++})} \cdot \frac{\mathcal{F}_{2^{++}}(N)^2}{\mathcal{F}_{0^{++}}(N)^2} = 2 \cdot \left(\frac{\mathcal{F}_{2^{++}}(N)}{\mathcal{F}_{0^{++}}(N)}\right)^2. \tag{10}$$

At $N=3$, the right-hand side equals 2 exactly. At $N=2$, the empirical ratio is $2.0014(17) \approx 2.00$, which by (10) requires $\mathcal{F}_{2^{++}}(2)/\mathcal{F}_{0^{++}}(2) \approx 1.0004$, i.e. $\mathcal{F}_{2^{++}}$ and $\mathcal{F}_{0^{++}}$ coincide at $N=2$ to four decimal places. We return to the N-dependence in §5.

4. PARI/GP verification of the rational Hecke eigenvalues

The Hecke eigenvalues (6) and (4) at $D=-67$ admit a direct algebraic computation using the ring of integers $\mathcal{O}_K = \mathbb{Z}[\alpha]$ where $\alpha = (1+\sqrt{-67})/2$, satisfying the minimal polynomial

$$\alpha^2 - \alpha + 17 = 0. \tag{11}$$

Lemma 4.1. *The smallest split prime of $K = \mathbb{Q}(\sqrt{-67})$ is $p = 17$.*

Proof. For each odd prime $p < 17$ with $p \neq 67$, we compute the Kronecker symbol $\chi_{-67}(p) = \left(\frac{-67}{p}\right)$ and check:

- $p=3$: $-67 \equiv 2 \pmod 3$, $\chi_{-67}(3) = (2/3) = -1$, inert.
- $p=5$: $-67 \equiv 3 \pmod 5$, $\chi_{-67}(5) = (3/5) = -1$, inert.
- $p=7$: $-67 \equiv 3 \pmod 7$, $\chi_{-67}(7) = (3/7) = -1$, inert.
- $p=11$: $-67 \equiv 10 \pmod{11}$, $\chi_{-67}(11) = (10/11) = -1$, inert.
- $p=13$: $-67 \equiv 11 \pmod{13}$, $\chi_{-67}(13) = (11/13) = -1$, inert.

The first p giving $\chi_{-67}(p) = +1$ is $p=17$: $-67 \equiv 1 \pmod{17}$. At $p=17$, the factorisation in $\mathcal{O}_K$ is $17 = \pi\bar{\pi}$ with $\pi = (1+\sqrt{-67})/2 = \alpha$, since $N(\alpha) = \alpha\bar{\alpha} = 17$ follows from (11). ☐

Proposition 4.2. *For $D=-67$, the Hecke eigenvalues at the smallest split prime $p=17$ are*

$$a_{17}(f_{-67}^{(3)}) = -33, \qquad a_{17}(f_{-67}^{(5)}) = +511. \tag{12}$$

Proof. By Lemma 4.1, $\pi = \alpha$ and $\pi^2 = \alpha^2 = \alpha - 17$ by (11). The trace map $\mathrm{Tr} : \mathcal{O}_K \to \mathbb{Z}$ gives:

$$\mathrm{Tr}(\pi^2) = \mathrm{Tr}(\alpha - 17) = \mathrm{Tr}(\alpha) - 34 = 1 - 34 = -33.$$

By (6), $a_{17}(f_{-67}^{(3)}) = \pi^2 + \bar{\pi}^2 = \mathrm{Tr}(\pi^2) = -33$.

For $a_{17}(f_{-67}^{(5)}) = \pi^4 + \bar{\pi}^4$, expand:

$$\pi^4 = (\pi^2)^2 = (\alpha - 17)^2 = \alpha^2 - 34\alpha + 289 = (\alpha - 17) - 34\alpha + 289 = -33\alpha + 272.$$

Then $\pi^4 + \bar{\pi}^4 = -33(\alpha + \bar{\alpha}) + 544 = -33 \cdot 1 + 544 = 511$, using $\alpha + \bar{\alpha} = \mathrm{Tr}(\alpha) = 1$.

Alternatively, the Newton identity (7) gives: $a_{17}(w=5) = (a_{17}(w=3))^2 - 2\cdot 17^2 = (-33)^2 - 578 = 1089 - 578 = 511$. $\square$

4.1. **PARI `mfeigenbasis` verification.** We verify Proposition 4.2 and extend the verification to the next three split primes $\{19, 23, 29\}$ using PARI/GP's `mfeigenbasis` function on the spaces $S_3^{\mathrm{new}}(67, \chi_{-67})$ and $S_5^{\mathrm{new}}(67, \chi_{-67})$.

The PARI session (full transcript reproducible at `phase_e1_pari_verify.gp` in the supplementary material) returns:

- $\dim S_3^{\mathrm{new}}(67, \chi_{-67}) = 13$, with 2 Galois-conjugate eigenforms: 1 $\mathbb{Q}$-rational eigenform (matching LMFDB `67.3.b.a`) and 1 Galois orbit of degree 10;
- $\dim S_5^{\mathrm{new}}(67, \chi_{-67}) = 23$, with 2 Galois-conjugate eigenforms: 1 $\mathbb{Q}$-rational eigenform (matching LMFDB `67.5.b.a`) and 1 Galois orbit of degree 20.

The number of $\mathbb{Q}$-rational eigenforms (1 at each weight) is consistent with HSH v3-OPUS, which predicts $\mathrm{rats}(D) = 2^{\mathrm{rk}_2(\mathrm{Cl}(K))} = 2^0 = 1$ at $D = -67$, $h_K = 1$ ([6]).

The first thirty coefficients of the $\mathbb{Q}$-rational weight-3 eigenform are:

$$[0, 1, 0, 0, 4, 0, 0, 0, 0, 9, 0, 0, 0, 0, 0, 0, 16, -33, 0, -29, 0, 0, 0, -21, 0, 25, 0, 0, 0, -9, 0],$$

and of the $\mathbb{Q}$-rational weight-5 eigenform:

$$[0, 1, 0, 0, 16, 0, 0, 0, 0, 81, 0, 0, 0, 0, 0, 0, 256, 511, 0, 119, 0, 0, 0, -617, 0, 625, 0, 0, 0, -1601, 0].$$

Vanishing at inert primes $\{2, 3, 5, 7, 11, 13, 31\}$ is consistent with the CM structure. The non-vanishing entries at $\{17, 19, 23, 29\}$ are the Hecke eigenvalues at the four smallest split primes.

Split p	$a_p(w=3)$ (PARI)	Predicted $a_p(w=5)$ via (7)	$a_p(w=5)$ (PARI)	Match
17	-33	$(-33)^2 - 578 = 511$	511	✓
19	-29	$(-29)^2 - 722 = 119$	119	✓
23	-21	$(-21)^2 - 1058 = -617$	-617	✓
29	-9	$(-9)^2 - 1682 = -1601$	-1601	✓

Table 1. Verification of the Newton identity (7) at the four smallest split primes of $K = \mathbb{Q}(\sqrt{-67})$.

Proposition 4.3. *The Newton identity* (7) *holds exactly at the four smallest split primes* $\{17, 19, 23, 29\}$ *of* $K = \mathbb{Q}(\sqrt{-67})$, *as verified by PARI/GP* `mfeigenbasis` *on the* $\mathbb{Q}$*-rational components of* $S_3^{\mathrm{new}}(67, \chi_{-67})$ *and* $S_5^{\mathrm{new}}(67, \chi_{-67})$.

Proof. Direct numerical verification in PARI/GP 2.15.4, see Table 4.1. The identity (7) is the elementary consequence of the explicit Newton power-sum identity at split primes; the numerical agreement at four distinct primes serves as a sanity check on the LMFDB labels `67.3.b.a` and `67.5.b.a` and on the choice of generator $\pi = (1 + \sqrt{-67})/2$ of the prime above $p = 17$. $\square$

5. Cross-N behaviour and Option B reframe

Athenodorou–Teper [1, Tables 34, 38] measures the ratio $r(N) := m_{2^{++}}(N)/m_{0^{++}}(N)$ for $N \in \{2, 3, 4, 5, 6, 8, 10, 12\}$ plus the $SU(\infty)$ continuum extrapolation. The data exhibit a monotone drift away from $\sqrt{2}$ for $N \geq 3$:

(13)

$$r(SU(2)) = 1.4147 \pm 0.0100 \quad (+0.04\%, \text{ Phase E core match}), \qquad r(SU(\infty)) = 1.4971 \pm 0.0082 \quad (+5.86\%, 10.1$$

A cross-N fit $r(N) = c_0 + c_1/N^2$ yields $c_0 = 1.489 \pm 0.007$, $c_1 = -0.32 \pm 0.05$, $\chi^2/\mathrm{dof} = 1.65$, with a large-$N$ asymptote 10.3σ off $\sqrt{2}$. The squared-ratio universality test $r(N)^2 = 2$ aggregates to $\chi^2 = 211.7$ on 9 d.o.f., equivalent to a 14.6σ rejection (reproducible code at `notes/AT2021_GEVP_refit_test.py`).

Interpretation. The $SU(2)$ "match" arises because the $1/N^2$ correction at $N = 2$ is $c_1/4 \approx -0.08$, reducing $r(2)$ from the large-N value 1.489 to $\approx 1.41 \simeq \sqrt{2}$ within the AT 2021 error. This is a finite-N coincidence, not a structural prediction. Of the nine data points, only $N = 2$ lies within 1σ of $\sqrt{2}$; seven of nine are outside 2.5σ.
Option B reframe. Per Plan v5 §C [10], the universal motivic identification $m^2_{2^{++}}/m^2_{0^{++}} = 2$ is falsified at 14.6σ aggregate (10.3σ in the direct c_0-fit). We retain Conjecture E-1 *only* as a single-discriminant, single-N empirical observation at $(D, N) = (-67, 2)$, at the Tier T_PLAUSIBLE level. Extension to other (D, N) requires a mechanism for the observed N-drift; the Casimir factor $\mathcal{F}_{2^{++}}(N)$ alone does not account for it.

Full cross-N tables, the lattice N-dependence analysis, the $c_{2^{++}} = 0.6385 \pm 0.0205$ extraction (8-point global fit), and the comparison of $\mathcal{F}_{0^{++}}(N)$ at the legacy $c = 0.80$ versus the P01 RELAUNCH update $c = 0.9446 \pm 0.0394$ are deferred to Appendix **??**.

An excited-state reframe is sketched in the companion Paper 6' (in preparation): the large-N data suggest the lattice-level *excited* 2^{++} state may match the AdS hard-wall Bessel prediction $j_{2,1}/j_{0,1} = 2.136$ at 2.1σ [2].

6. Additive decomposition Conjecture E-3

The empirical $SU(3)$ J^{PC} spectrum of [1, Table 34], divided by $m^2(0^{++})$, suggests an additive structure:

Conjecture E-3 (additive J^{PC} decomposition)**.** *For the heterotic-CM-*K3 *family at $D = -67$, $SU(3)$, the J^{PC} glueball spectrum admits the decomposition*

$$\frac{m^2(X^{J^{PC}})}{m^2(0^{++})} = 1 + a_J \cdot \mathbb{1}_{J=2} + a_P^- \cdot \mathbb{1}_{P=-} + a_C^- \cdot \mathbb{1}_{C=-} + \delta_X, \tag{14}$$

with empirical coefficients $a_{J=2} \approx 1.07$, $a_P^- \approx 1.40$, $a_C^- \approx 2.16$, and small residuals $|\delta_X| \lesssim 0.05$.

Testing against the $SU(3)$ data:

$X^{J^{PC}}$	$m^2/m^2_{0^{++}}$ AT 2021	E-3 prediction	Residual δ_X
0^{++}	1.000	1 (anchor)	0
2^{++}	2.066	$1 + 1.07 = 2.07$	-0.004
0^{-+}	2.400	$1 + 1.40 = 2.40$	0 (anchor)
2^{-+}	3.452	$1 + 1.07 + 1.40 = 3.47$	-0.018
1^{+-}	3.173	$1 + a_{J=1} + 2.16$	-0.01 (fit $a_{J=1}$)
2^{--}	5.626	$1 + 1.07 + 1.40 + 2.16 = 5.63$	-0.004

The residuals are at the $\sim 0.5\%$ level, comparable to the lattice errors ($\sim 1\%$ on the squared ratios). The coincidence at 2^{-+} and 2^{--} is nontrivial: each is a sum of independent additive contributions calibrated elsewhere, with no parameter freedom.

Remark 6.1. Conjecture E-3 is empirical at the present stage and is at the T_SPECULATIVE tier. It motivates a search for an arithmetic decomposition of the J^{PC} spectrum into CM newform contributions; we identify candidates in §9.

7. Falsifiability

Conjecture E-1 admits a sharp falsifiability test that is feasible with current lattice technology. We isolate the most discriminating prediction:

Proposition 7.1 (falsifiability test)**.** *A high-precision continuum-extrapolated lattice measurement of $(m(2^{++})/m(0^{++}))^2$ at $SU(2)$ outside the interval $[1.99, 2.01]$ falsifies Conjecture E-1 in its present form.*

Proof. By (10) with $\mathcal{F}_{2^{++}}(2)/\mathcal{F}_{0^{++}}(2) = 1$ (the requirement of §3.1), the prediction is $2 \pm \epsilon$ with ϵ controlled by the uncertainty on the higher-order $1/N^4$ corrections, which the AT 2021 fit at the SOTA precision 0.4% constrains to $\epsilon \lesssim 0.01$ at $N = 2$. □

The current AT 2021 value is 2.001 ± 0.029; a factor-5 reduction in the error bar (achievable with a doubled-basis GEVP and $\sim 5 \times 10^4$ configurations) would shrink the uncertainty to ~ 0.003, fully resolving the test.

7.1. **Secondary tests.**

- The value of $c_{2^{++}}$ extracted by fitting the SU(N) 2^{++} ground state to $\mathcal{F}_{2^{++}}(N) = (1+c_{2^{++}}/N^2)/(1+c_{2^{++}}/9)$ should fall in $[0.65, 0.80]$. Outside this range, the monotonicity conjecture (15) is in question.
- The additive decomposition Conjecture E-3 predicts $m^2(2^{--})/m^2(0^{++}) \approx 5.63$. The AT 2021 value at $SU(3)$ is 5.626, leaving residual $\sim 0.07\%$. The same prediction at $SU(2)$ and $SU(8)$ provides further tests.
- The pseudoscalar mass ratio $(m_{0^{-+}}/m_{0^{++}})^2_{SU(2)} = 2.531(33)$ is at $w_{\text{mot}}(0^{-+}) = 2.531 \times 2 - 0 = 5.06$, not an integer. This indicates that the 0^{-+} is not directly identified with a single CM newform weight in the proposed assignment; it likely couples to topological structure via the Witten–Veneziano sum rule [17] (open, T_SPECULATIVE).

8. Open problems

8.1. **The $\mathbf{0^{-+}}$ pseudoscalar and topological susceptibility.** The 0^{-+} pseudoscalar glueball mass squared satisfies, in pure SU(N) lattice [1, Table 35], $(m_{0^{-+}}/m_{0^{++}})^2_{SU(N)}$ approximately constant at ~ 2.4 across N. The Witten–Veneziano sum rule [17, 15] relates the pseudoscalar mass to the topological susceptibility χ_t:

$$m^2_{0^{-+}}\, f^2_{gb} \;\sim\; \frac{\chi_t}{N^2-1}, \tag{15}$$

with f_{gb} a glueball-sector decay constant analogous to f_π. The ECI v15 framework on K3$_{-67}$ admits two structural candidates for χ_t:

1. The Euler characteristic $\chi(\mathrm{K3}) = 24$ as a topological invariant, modulated by the period $\langle \Omega, \bar{\Omega} \rangle$;
2. The L-value $L(f^{(3)}_{-67}, 1)$ at the leftmost critical point of the weight-3 CM newform L-function, by Damerell–Yager [18] machinery.

We do not pursue this further here.

8.2. **The glueball mass-radius.** Shanahan et al. [11] report the first lattice computation of the $SU(3)$ scalar glueball gravitational form factor, with mass-radius $r_{\text{mass}} = 0.263 \pm 0.031$ fm. The naive Compton wavelength $\hbar c/m_{0^{++}} = 0.116$ fm is a factor 2.3 smaller. Within the ECI v15 framework, a natural candidate for the squared radius is

$$r_g^2(SU(N)) \;\sim\; \frac{\langle \Omega, \bar{\Omega} \rangle_{\mathrm{K3}_{-67}}}{m^2_{0^{++}}(SU(N)) \cdot (N^2-1)} \cdot \kappa, \tag{16}$$

with κ an order-unity constant to be fixed. The Petersson norm $\langle \Omega, \bar{\Omega} \rangle$ on a CM-K3 is computable from Chowla–Selberg-type formulae for the period of the weight-3 form. We leave the explicit evaluation of (19) for future work.

8.3. **Extension to other Heegner discriminants.** The construction of Section 2.2 extends to all five remaining $h_K = 1$ Heegner discriminants $D \in \{-7, -11, -19, -43, -163\}$, with LMFDB labels D.5.b.a in each case. Since the ECI v15 anchor specifically singles out $D = -67$ (via the four selection criteria of [4, §1.3]), the question of whether Conjecture E-1 holds at other D as a tautological identity (no Yang–Mills theory associated) or as a non-trivial prediction (multiple choices of anchor possible) is structurally important.

8.4. **From conjecture to derivation.** The principal open problem is the derivation of Conjecture E-1 from the heterotic-CM-K3 spectral structure. Three candidate routes:

(1) *Spectral triple route*: extending the conditional eigenvalue identification of [3, §6] (under hypotheses (S), (C)) from the scalar to higher-spin sectors. The Hodge decomposition of $H^2(\mathrm{K3})$ into transcendental and algebraic parts may be the natural index for J^{PC} assignment.
(2) *Kuga–Sato fourfold lift route*: the Hodge_fourfolds construction [5, §2.3] embeds $\rho_{f^{(5)}_{-67}}$ in $H^4((E_{-67})^4)$ as the extreme $\{\pi^4, \bar{\pi}^4\}$ eigenvalue sub-representation of $\mathrm{Sym}^4(H^1(E_{-67}))$. The five-dimensional $\mathrm{Sym}^4(H^1(E_{-67}))$ admits a natural physical interpretation as the multiplet of spin-2 modes.
(3) *AdS/CFT analogue route*: in the holographic dictionary, glueballs are eigenmodes of the bulk Laplacian on the warped throat geometry. The arithmetic heterotic-CM-K3 construction may provide an arithmetic refinement of the AdS/CFT glueball spectrum of [2, Section 4] predicting J^{PC} ratios in agreement with our empirical Conjecture E-1.

9. Conclusion

We have isolated a numerical pattern in the Athenodorou–Teper 2021 continuum lattice spectrum of pure $SU(2)$ Yang–Mills theory: $(m_{2^{++}}/m_{0^{++}})^2 = 2.001 \pm 0.029$ at $SU(2)$, matching the integer $2 = 4/2$ to within 0.08σ. We have proposed the natural arithmetic interpretation of this integer as the ratio of motivic weights $w_{\mathrm{mot}}(f^{(5)}_{-67})/w_{\mathrm{mot}}(f^{(3)}_{-67}) = 4/2$ of two canonical CM newforms attached to $K = \mathbb{Q}(\sqrt{-67})$, and formulated this as Conjecture E-1. The arithmetic foundation (rational Hecke eigenvalues at $\{17, 19, 23, 29\}$, Newton identity (7)) is verified by PARI/GP.

Cross-N limitation (added post-draft, 2026-05-15). A cross-N audit (Section 6) establishes that the large-N extrapolation of $(m_{2^{++}}/m_{0^{++}})$ is $c_0 = 1.489 \pm 0.007$, discrepant from $\sqrt{2}$ at 10.3σ (direct c_0 fit). The aggregate cross-N chi-squared falsification of the universal ratio $r(N) \equiv 2$ is 14.6σ (naive $\sqrt{\chi^2}$; Gaussian-equivalent $z = 13.4\sigma$, $p \sim 10^{-40}$). The $SU(2)$ match is a finite-N coincidence. **Conjecture E-1 as a universal cross-N law is falsified.** It is retained as a single-discriminant, single-N empirical observation at $(D, N) = (-67, 2)$, tier T_PLAUSIBLE, not as a structural prediction.

The Phase E motivic anchor remains valid as an empirical $SU(2)$-specific observation. Cross-N universality requires either a new theoretical mechanism or a fundamentally different observable. The companion Paper 6' (in preparation) explores the excited-state AdS hard-wall reframe, where two large-N Bessel matches survive at $< 2\sigma$: $0^{-+}_{\mathrm{ex1}}/0^{++}_{\mathrm{gs}} \approx j_{0,2}/j_{0,1}$ (0.32σ) and $2^{++}_{\mathrm{ex1}}/0^{++}_{\mathrm{gs}} \approx j_{2,1}/j_{0,1}$ (1.34σ) [2].

A tentative additive decomposition Conjecture E-3 fits the $SU(3)$ J^{PC} spectrum at the $\sim 0.5\%$ level using three empirical parameters; it is at the T_SPECULATIVE tier and requires independent confirmation.

The present paper is targeted at the *Journal of Number Theory* as a short note documenting the arithmetic observation and its finite-N limitations, with explicit falsifiability conditions.

Appendix A. Cross-N audit details

This appendix collects the full cross-N data and analysis underlying the falsification statement of Section 6.

A.1. **Drift table for $(m_{2^{++}}/m_{0^{++}})^2$.** Athenodorou–Teper [1, Table 34]:

N	$(m_{2^{++}}/m_{0^{++}})^2$	Drift vs. 2
2	2.001 ± 0.029	0.07%
3	2.066 ± 0.026	3.3%
4	2.102 ± 0.034	5.1%
5	2.208 ± 0.041	10.4%
6	2.274 ± 0.052	13.7%
8	2.261 ± 0.040	13.0%

Under Conjecture E-1, the drift implies $\mathcal{F}_{2^{++}}(N) \neq \mathcal{F}_{0^{++}}(N)$ for $N \geq 3$ via

$$\frac{\mathcal{F}_{2^{++}}(N)}{\mathcal{F}_{0^{++}}(N)} = \sqrt{\frac{1}{2}\,\frac{m^2(2^{++}, SU(N))}{m^2(0^{++}, SU(N))}}. \tag{17}$$

A.2. **Extraction of $c_{2^{++}}$.** If $\mathcal{F}_{2^{++}}(N) = (1 + c_{2^{++}}/N^2)/(1 + c_{2^{++}}/9)$ with anchor $\mathcal{F}_{2^{++}}(3) = 1$, the SOTA $N = 2$ ratio $5.349/4.894 = 1.0930 \pm 0.005$ implies $c_{2^{++}} = 0.72 \pm 0.04$ (2-anchor local fit). The $N = 2, \ldots, 12$ global fit on AT 2021 Tables 34–35 (`notes/AT2021_GEVP_refit_ConjE1_retest_2026-05-16` §2.3) yields $c_{2^{++}} = 0.6385 \pm 0.0205$, in 3.97σ tension with the 2-anchor extraction; the discrepancy is consistent with non-monotone sub-leading $1/N^4$ contamination at large N. The 0^{++} value is $c_{0^{++}} = 0.9446 \pm 0.0394$ (P01 RELAUNCH 2026-05-16, superseding the legacy $c_{0^{++}} = 0.80 \pm 0.05$ of [13]).

Remark A.1. The decrease of c_X as spin increases is physically plausible: the 't Hooft $1/N^2$ correction is suppressed for higher-spin states because planar-diagram resummation is dominated by larger-genus contributions when more derivatives are involved. A first-principles derivation of c_X from the heterotic CM-K3 structure remains open; we conjecture monotonicity $c_{0^{++}} > c_{2^{++}} > c_{0^{-+}} > c_{2^{-+}} > \cdots$ as a derived prediction of Conjecture E-1 at Tier T_PLAUSIBLE.

A.3. **Legacy comparison of $\mathcal{F}_{0^{++}}(N)$ with $c = 0.80$.** For traceability of the calibration timeline (cf. post-2026-05-16 update, $c = 0.94 \pm 0.04$ now favoured at $\chi^2/\mathrm{ndf} = 1.49$, $p = 0.165$, with $c = 0.80$ disfavoured at 3.67σ; see `notes/F_N_revisit_2026-05-16.py`):

N	ECI v15 ($c = 0.80$)	AT 2021	Tension
2	3.752	3.781(23)	1.3σ
3	3.405	3.405(21)	anchor
4	3.284	3.271(27)	0.5σ
5	3.226	3.156(31)	2.3σ
6	3.197	3.102(32)	3.0σ
8	3.166	3.099(26)	2.6σ
∞	3.127	~ 3.07	$\sim 0.8\sigma$

The legacy fit is excellent at $N \in \{2, 3, 4\}$ ($\leq 1.3\sigma$); the tension grows to $\sim 3\sigma$ at $N \in \{5, 6, 8\}$, motivating the P01 RELAUNCH recalibration to $c = 0.9446 \pm 0.0394$.

A.4. **Full $r(N)$ cross-N table.** The ratio $r(N) := m_{2^{++}}(N)/m_{0^{++}}(N)$ on AT 2021 Tables 34–38:

N	$r(2^{++}/0^{++})$	Deviation from $\sqrt{2} = 1.4142$
2	1.4147(100)	+0.04% — Phase E core match
3	1.4373(97)	+1.63%
4	1.4497(129)	+2.50%
5	1.4861(157)	+5.06%
6	1.5081(169)	+6.62%
8	1.5037(140)	+6.32%
10	1.4810(188)	+4.71%
12	1.4745(171)	+4.26%
∞ (Table 38)	$\mathbf{1.4971 \pm 0.0082}$	+5.86%, 10.1σ off $\sqrt{2}$
fit $c_0 + c_1/N^2$	$\mathbf{1.489 \pm 0.007}$	+5.29%, 10.3σ via fit

The aggregate squared-ratio test $r(N)^2 = 2$ gives $\chi^2 = 211.7$ on 9 d.o.f., equivalent to a 14.6σ rejection of the universal motivic identification (naive $\sqrt{\chi^2}$ convention; the Gaussian-equivalent two-sided $z = 13.4\sigma$, $p \sim 10^{-40}$).

Acknowledgments

This work builds on the ECI v15 closed-form mass-gap framework of [13], the canonical weight-5 CM newform construction of [5], and the continuum-extrapolated lattice $SU(N)$ glueball spectrum of [1]. We thank the LMFDB collaboration for the public availability of the Hecke eigenvalue tables that enabled the present verification of the rational Hecke coefficients at `67.3.b.a` and `67.5.b.a`.

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